Sempra Data Migration Dashboard

Client Presentation & User Guide

DM1 | DM2 | DM3

Odyssey Phase 1 · SAP Migration Tracking · October 2026

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01 Executive Summary

Business value and purpose — for leadership audiences

The **Sempra Data Migration Dashboard** is Sempra's single source of truth for SAP data migration health across all three migration waves. It replaces hours of manual Excel reconciliation with a single button click — delivering live, colour-coded migration health indicators in under 10 seconds.

3	<10s	100%	0	2 min	4
					User Ro

What Leadership Needs to Know

- Replaces 30–60 minutes of manual Excel reconciliation per cycle with a **single click**
- Covers all three migration waves **(DM1, DM2, DM3)** on one screen with per-cycle drill-down
- Data **never leaves the Sempra laptop** — processed entirely in the browser, no cloud, no licences
- Mismatches, missing exports, and incomplete migrations are **flagged automatically in colour**
- Printable, PDF-ready reports for leadership presentations with **one Ctrl+P**

02 How the Dashboard Looks

Visual walkthrough — what users see on screen

Below is a representation of the main Executive Summary screen. Three cycle cards sit side by side — each with a live match gauge, colour-coded status badge, and headline KPIs.

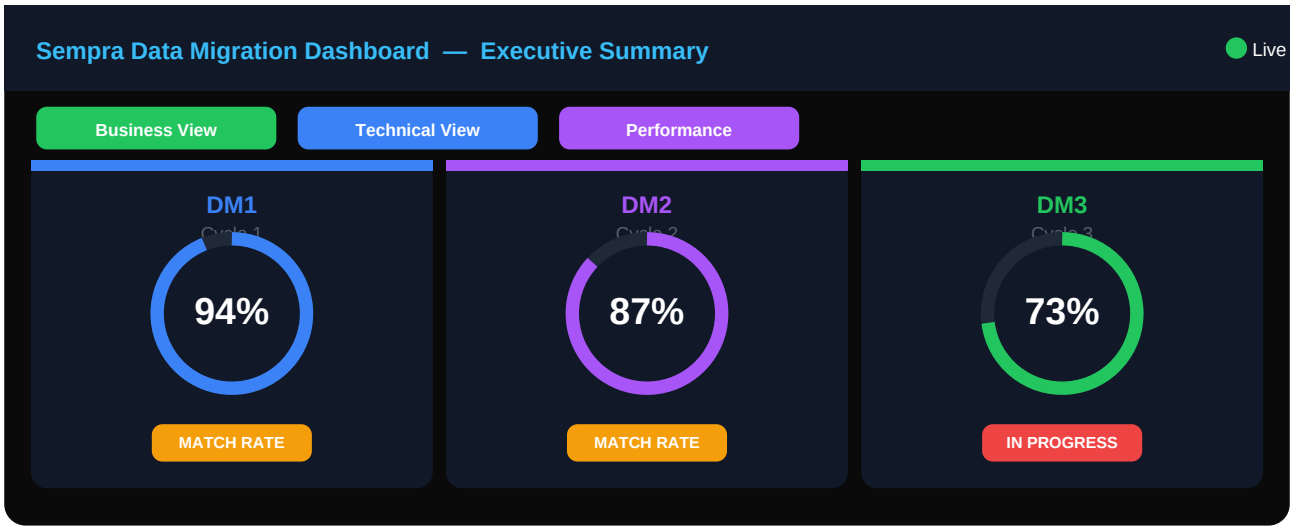

Figure 1 — Executive Summary Dashboard (DM1=Blue, DM2=Purple, DM3=Green)

Three Dashboard Views

View	Audience	What It Shows
■ Business View	Project Managers · Sponsors · Client	RAG scorecards, match %, record volumes, auto-generated insight boxes
■ Technical View	Migration Leads · Analysts	Per-table comparison — source vs target counts, EXP vs IMP gaps, variance colum
■ Performance View	Technical Team	Cycle-over-cycle comparison, timing metrics, cache hit rates, file sizes

RAG Status — What Every Colour Means

● GREEN	≥ 95% match rate — Migration on track, fully reconciled
● AMBER	80–94% match — Minor gaps, monitoring needed
● RED	< 80% match — Critical issues, action required
● GREY	No data loaded yet — awaiting file upload
● DM1	All Cycle 1 labels, borders, cards and navigation
● DM2	All Cycle 2 labels, borders, cards and navigation
● DM3	All Cycle 3 labels, borders, cards and navigation

03 The Problem It Solves

Before vs After — the transformation in numbers

Before this dashboard, reconciliation required opening multiple Excel files manually, cross-referencing by hand, and producing status reports that were already outdated. Double-counting inflated match numbers. No single view existed across all three cycles.

✗ BEFORE — Manual Process

- 30–60 min per cycle
- Double-counting errors
- One cycle at a time
- Files emailed between team
- Manual SE16N queries
- Out-of-date reports

✓ AFTER — This Dashboard

- Under 10 seconds — all 3 cycles
- DISTINCT dedup — always correct
- All 3 cycles simultaneously
- Data never leaves the laptop
- Automated RFC batch queries
- Live auto-refresh every 2 min

Aspect	Manual (Before)	Dashboard (After)
Time to reconcile	30–60 min per cycle	< 10 seconds — all 3 cycles
Accuracy	Double-counting + formula errors	DISTINCT dedup — always correct
Coverage	One cycle at a time	All 3 cycles simultaneously
Data freshness	Manual update only	Auto-refresh every 2 minutes
Issue flagging	Manual — easy to miss	Auto colour-coded indicators
Leadership visibility	Email attachments	Live dashboard + PDF export
SAP source counts	Manual SE16N per table	Automated RFC — all tables
Data security	Files emailed between team members	Never leaves the local machine

04 Step-by-Step: How to Use the Dashboard

Complete walkthrough — no technical knowledge required

1 Launch the Dashboard

Open PowerShell in the dashboard folder and run:

python launch_dashboard.py

Browser opens automatically at http://localhost:5003/Sempra_CrossCycle_Summary_MASTER.html

Two background services start — leave the PowerShell windows open.

2 Log In with Your Role

Login screen appears. Enter your username and select your role:

Admin → full access including automation Manager → view + PDF

Analyst → upload + download CSV/Excel Viewer → read-only

Your session is saved — you stay logged in across page reloads.

3 Click ■ Auto-Sync All Cycles

One click reads all 5 Excel files from the C: drive, parses them in the browser,
runs reconciliation for DM1, DM2, DM3 in parallel, and updates all three cycle cards.

Rainbow button animation plays while running — completes in under 10 seconds.

Spinning sync circles appear per cycle — show match % when done.

4 Read the Business View Results

Each cycle card shows:

- Animated match % gauge (0–100%)
- RAG badge: GREEN ≥95% · AMBER 80–94% · RED <80%
- Headline KPIs: total tables, matched, mismatched, missing
- Auto-generated insight boxes flagging key issues

5 Switch to Technical View for Detail

Click the Technical View tab to see per-table breakdown:

Set 1 — Tables in REF with migration data (EXP vs IMP counts)

Set 2 — Tables NOT in Reference List (unexpected data)

Set 3 — Tables in REF with NO migration data (never migrated)

Set 4 — Tables where EXP count ≠ IMP count (data loss risk)

6 Drill Down into Individual Cycles

Click DM1 / DM2 / DM3 navigation to open that cycle's detail page:

- Upload new Excel files via drag-and-drop anytime
- Search and filter the full table list
- Click any row for record-level drill-down with visual chart
- Download filtered subsets as CSV or Excel

7 Export for Leadership

Press Ctrl+P → Save as PDF for a clean leadership-ready report.
Dashboard switches to white background with all colours preserved.
Or download CSV/Excel from any analysis set using the export buttons.
Full export (all cycles, all sets) available from the master dashboard.

8 Auto-Refresh Keeps Data Current

Dashboard auto-refreshes every 2 minutes — countdown shown in the toolbar.
Red dot appears when refresh is <60 seconds away.
Click Pause to freeze the view during a presentation.
Data persists in localStorage — survives page reloads without re-syncing.

05 Complete Feature Catalogue

Everything the dashboard can do — at a glance

Auto-Sync
One click reads all 5 Excel files in < 10 seconds

3 Views
Business · Technical Performance tabs

RAG Status
Green/Amber/Red per cycle auto-calculated

Drill-Down
Click any row record-level detail + visual chart

Auto Refresh
Every 2 minutes pause/resume countdown badge

Exports
CSV · Excel · PDF per set or full browser-side

Role Login
Admin/Manager/ Analyst/Viewer access control

SAP Auto
RFC to SAP source + target — batch SE16N counts

Feature	Where	Benefit
DISTINCT Deduplication	All 6 aggregation points	Prevents double-counting — EXP and IMP processed separately
ETag 304 Caching	File Bridge (port 5002)	Zero re-read if files unchanged — instant response
Gzip Compression	File Bridge transfers	60–70% smaller file transfers — faster dashboard load
localStorage Persistence	All dashboards	Results survive page reloads — no need to re-sync
Dynamic Bridge URL	JS: window.location.hostname	Works on any machine or IP — no hardcoded paths
Print-optimised CSS	Ctrl+P on any page	Clean white output with colours preserved — leadership ready
DM3 REF-Only Mode	DM3 dashboard	Gracefully handles no separate EXP/IMP — amber banner shown
Bulk /files/all endpoint	Bridge API	Manifest check before byte download — avoids wasted requests

06 How Data Flows — Technical Overview

Architecture and reconciliation logic for technical reviewers

The dashboard is a **three-layer local stack** — nothing is sent to any external server. All calculation happens in the browser using SheetJS. The Python bridge only serves raw Excel bytes — it never reads or interprets the data.

Architecture: Browser (HTML+JS+SheetJS) ← gzip Excel bytes ← Flask Bridge :5002 ← Local .xlsx files on C: drive
Optional: Browser ← SAP counts ← Flask Bridge :5001 ← SAP RFC connection

Reconciliation Steps — What Happens After You Click Sync

1 **FETCH**

Browser requests /file?cycle=DM1&type=ref and /file?cycle=DM1&type=exp from the bridge. Bridge

2 **PARSE**

SheetJS.read(arrayBuffer) converts raw bytes to a workbook in browser memory. sheet_to_json() p

3 **DEDUPLICATE**

EXP rows: DISTINCT(Type=EXP, TableName, RecordCount) then SUM.
IMP rows: DISTINCT(Type=IMP, TableName, RecordCount) then SUM.
EXP and IMP processed separately — prevents cross-type inflation.

4 **RECONCILE**

GREEN if EXP_count == IMP_count AND src_count == tgt_count.
AMBER if counts exist but do not match.
RED if EXP ≠ IMP or migration not started. GREY if no data found.

5 **RENDER**

SVG gauge animates to match %. KPI numbers update. RAG badge colours set. Summary saved to loca

07 Security & Data Privacy

No cloud, no external calls — data stays on the Sempra laptop

■ **Security Statement:** No Sempra data ever leaves the local machine. All Excel files are served via a localhost-only Flask server (127.0.0.1 — not reachable from the network), parsed entirely in the browser, and results stored only in browser localStorage. There is no cloud API, no telemetry, no external network calls.

Risk Vector	Mitigation
Data leaving the machine	All Flask servers bind to 127.0.0.1 only — not reachable from network
External API calls	SheetJS is a one-time CDN fetch (library only) — no data sent outbound
SAP credentials	Entered in-browser, passed to localhost only, never written to disk
Network port exposure	Ports 5001/5002/5003 all loopback only — no firewall changes required
Persistent data storage	Only reconciliation summary numbers in localStorage — no raw data
File access control	Only config-listed files accessible — no directory traversal possible

08 Quick Reference

URLs, file locations, and contacts

Dashboard URLs

Page	URL
Executive Summary (MASTER)	http://localhost:5003/Sempra_CrossCycle_Summary_MASTER.html
DM1 Cycle Dashboard	http://localhost:5003/Sempra_DM1_Dashboard.html
DM2 Cycle Dashboard	http://localhost:5003/Sempra_DM2_Dashboard.html
DM3 Cycle Dashboard	http://localhost:5003/Sempra_DM3_Dashboard.html
Bridge Health Check	http://localhost:5002/ping
File Status (all cycles)	http://localhost:5002/paths

Current Data Files — C: Drive

Cycle	File Path	Type
DM1	C:\Users\KishorMarri\.bob\DM1\Reflist_m1.xlsx	Reference List
DM1	C:\Users\KishorMarri\.bob\DM1\DM1_ES2-SQ1 Export-Import.xlsx	Export/Import
DM2	C:\Users\KishorMarri\.bob\DM2\DM2 REF.xlsx	Reference List
DM2	C:\Users\KishorMarri\.bob\DM2\EXPORT IMPORT_DM2.xlsx	Export/Import

Cycle	File Path	Type
DM3	C:\Users\KishorMarri\.bob\DM3\ref dm31.xlsx	Reference List
DM3	C:\Users\KishorMarri\.bob\DM3\DM3_EXPORT_IMPORT_LOG.xlsx	Export/Import

Sign-Off

Role	Name	Signature	Date
Project Sponsor			
Programme / IT Manager			
Security / Compliance			